

1. Parts (a) and (b) are not related.

- (a) (6 points) Find parametric equations for the line through the point $(0, 3, 2)$ that is parallel to the plane $x + y + z = 1$ and perpendicular to the line $x = t, y = 3 - t, z = 2 + 2t$.

Direction vector of line is orthogonal to:

• Normal vector of $x + y + z = 1$

• Direction vector of $x = t, y = 3 - t, z = 2 + 2t$

Direction is $\langle 1, 1, 1 \rangle \times \langle 1, -1, 2 \rangle = \langle 3, -1, -2 \rangle$

$$\begin{aligned} x &= 3t \\ y &= 3 - t \\ z &= 2 - 2t \end{aligned}$$

Parametric equations: _____

$$\begin{aligned} x &= 3t \\ y &= 3 - t \\ z &= 2 - 2t \end{aligned}$$

- (b) (1.5 points each) Consider the quadric surface $x^2 + 2z^2 = y^2$. For each of the following planes, identify the trace of the quadric surface in that plane.

Circle one answer for each part. You do not have to show any work.

- i. $x = 1$:
 point line pair of lines circle
 ellipse parabola hyperbola does not exist

$$1 + 2z^2 = y^2$$

- ii. $y = 1$:
 point line pair of lines circle
ellipse parabola hyperbola does not exist

$$x^2 + 2z^2 = 1$$

- iii. $z = 0$:
 point line pair of lines circle
 ellipse parabola hyperbola does not exist

$$x^2 = y^2$$

- iv. $y = 0$:
point line pair of lines circle
 ellipse parabola hyperbola does not exist

$$x^2 + 2z^2 = 0$$

2. (12 points) Find the equation of the tangent plane to the surface S at the point $P(4, -1, 5)$ given that the two curves

$$\mathbf{r}_1(t) = \langle 2t^2 + 3t + 5, 4t + 3, 4 - t^3 \rangle$$

and

$$\mathbf{r}_2(s) = \left\langle s^2, \frac{2}{s} - s, 2s + 1 \right\rangle$$

are on the surface S and they intersect at the point $P(4, -1, 5)$.

Give your answer in standard form: $Ax + By + Cz = D$.

$$\vec{r}_1(t) = \langle 4, -1, 5 \rangle \text{ at } t = -1$$

$$\vec{r}_2(s) = \langle 4, -1, 5 \rangle \text{ at } s = 2$$

Tangent vectors are both in tangent plane:

$$\vec{r}_1'(t) = \langle 4t + 3, 4, -3t^2 \rangle \quad \vec{r}_1'(-1) = \langle -1, 4, -3 \rangle$$

$$\vec{r}_2'(s) = \langle 2s, -\frac{2}{s^2} - 1, 2 \rangle \quad \vec{r}_2'(2) = \langle 4, -\frac{3}{2}, 2 \rangle$$

$$\text{Normal vector: } \langle -1, 4, -3 \rangle \times \langle 4, -\frac{3}{2}, 2 \rangle = \left\langle \frac{7}{2}, -10, -\frac{29}{2} \right\rangle$$

$$\frac{7}{2}(x-4) - 10(y+1) - \frac{29}{2}(z-5) = 0$$

$$7(x-4) - 20(y+1) - 29(z-5) = 0$$

$$7x - 20y - 29z = -97$$

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Tangent plane (in standard form):

3. (12 points) Find the linear approximation for

$$g(x, y, z) = 4\sqrt{x^2 + y^2} + 5\sqrt{4y^2 + z^2}$$

at the point $(3, 4, 6)$ and use it to approximate the value $g(2.97, 4.01, 6.04)$. Round your answer to three digits after the decimal.

$$g_x(x, y, z) = \frac{4x}{\sqrt{x^2 + y^2}}$$

$$g_x(3, 4, 6) = 2.4$$

$$g_y(x, y, z) = \frac{4y}{\sqrt{x^2 + y^2}} + \frac{20y}{\sqrt{4y^2 + z^2}}$$

$$g_y(3, 4, 6) = 11.2$$

$$g_z(x, y, z) = \frac{5z}{\sqrt{4y^2 + z^2}}$$

$$g_z(3, 4, 6) = 3$$

$$g(3, 4, 6) = 70$$

$$L(x, y, z) = 2.4(x - 3) + 11.2(y - 4) + 3(z - 6) + 70$$

$$g(2.97, 4.01, 6.04) \approx L(2.97, 4.01, 6.04)$$

$$= 2.4(-0.03) + 11.2(0.01) + 3(0.04) + 70$$

$$= 70.16$$

$$L(x, y, z) =$$

$$2.4(x - 3) + 11.2(y - 4) + 3(z - 6) + 70$$

$$g(2.97, 4.01, 6.04) \approx 70.16$$

4. (12 points) Find and classify all critical points for the function

$$f(x, y) = -\frac{1}{4}x^4 + \frac{2}{3}x^3 + 4xy - y^2.$$

$$f_x(x, y) = -x^3 + 2x^2 + 4y = 0$$

$$f_y(x, y) = 4x - 2y = 0 \rightarrow y = 2x$$

$$-x^3 + 2x^2 + 8x = 0$$

$$-x(x^2 - 2x - 8) = 0$$

$$-x(x-4)(x+2) = 0$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ x=0 & x=4 & x=-2 \\ y=0 & y=8 & y=-4 \end{array}$$

3 crn. pts: $(0,0)$, $(4,8)$, $(-2,-4)$

$$f_{xx}(x, y) = -3x^2 + 4x$$

$$f_{yy}(x, y) = -2$$

$$f_{xy}(x, y) = 4$$

$$D(0,0) = 0 - 16 < 0$$

$\rightarrow$ saddle pt

$$D(4,8) = 64 - 16 > 0, \quad f_{xx}, f_{yy} < 0 \rightarrow \text{local max}$$

$$D(-2,-4) = 40 - 16 > 0, \quad f_{xx}, f_{yy} < 0 \rightarrow \text{local max}$$

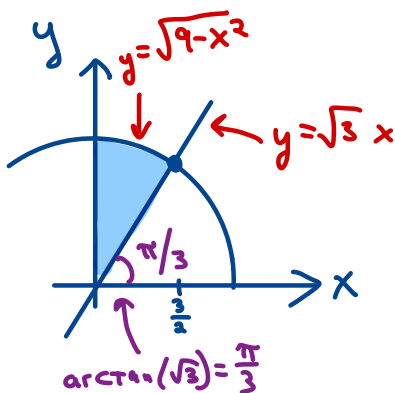
saddle pt @ $(0,0)$

local max @ $(4,8)$

local max @ $(-2,-4)$

Points and classification:

5. (12 points) Evaluate the integral by converting to polar coordinates:



$$\int_0^{3/2} \int_{(\sqrt{3})x}^{\sqrt{9-x^2}} 2xy \, dy \, dx$$

$\downarrow$
 $2r \cos \theta \, r \sin \theta$

$$\int_{\pi/3}^{2\pi/3} \int_0^3 2r^3 \cos \theta \sin \theta \, dr \, d\theta$$

$$= \int_{\pi/3}^{2\pi/3} \cos \theta \sin \theta \left(\frac{1}{2} r^4 \right) \Big|_{r=0}^{r=3} d\theta$$

$$= \int_{\pi/3}^{2\pi/3} \frac{81}{2} \sin \theta \cos \theta \, d\theta$$

$u = \sin \theta$
 $du = \cos \theta \, d\theta$

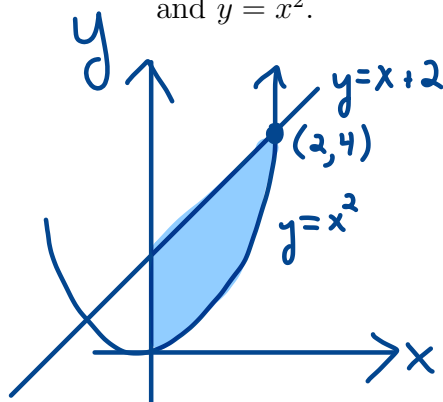
$$= \int_{\frac{\sqrt{3}}{2}}^1 \frac{81}{2} u \, du = \frac{81}{4} u^2 \Big|_{\frac{\sqrt{3}}{2}}^1$$

$$= \frac{81}{4} \left(1 - \frac{3}{4} \right) = \frac{81}{16}$$

$$\boxed{\frac{81}{16}}$$

Answer: _____

6. (12 points) Find the y -coordinate $\bar{y}$ of the center of mass of the lamina with density function $\rho(x, y) = x$ that occupies the region D in the first quadrant bounded by the curves $y = x + 2$ and $y = x^2$.



$$m = \int_0^2 \int_{x^2}^{x+2} x \, dy \, dx = \int_0^2 \left(yx \right)_{y=x^2}^{y=x+2} dx = \int_0^2 (x(x+2) - x^3) dx$$

$$= \int_0^2 (x^2 + 2x - x^3) dx = \left(\frac{1}{3}x^3 + x^2 - \frac{1}{4}x^4 \right) \Big|_0^2 = \frac{8}{3} + 4 - 4 = \frac{8}{3}$$

$$M_x = \int_0^2 \int_{x^2}^{x+2} yx \, dy \, dx = \int_0^2 \left(\frac{1}{2}y^2x \right)_{y=x^2}^{y=x+2} dx = \int_0^2 \left(\frac{1}{2}(x+2)^2x - \frac{1}{2}x^5 \right) dx$$

$$= \frac{1}{2} \int_0^2 (x^3 + 4x^2 + 4x - x^5) dx$$

$$= \frac{1}{2} \left(\frac{1}{4}x^4 + \frac{4}{3}x^3 + 2x^2 - \frac{1}{6}x^6 \right) \Big|_0^2$$

$$= \frac{1}{2} \left(4 + \frac{32}{3} + 8 - \frac{64}{6} \right) = 6$$

$$\bar{y} = \frac{M_x}{m} = \frac{6}{(\frac{8}{3})} = \frac{9}{4} = 2.25$$

$$\bar{y} = \boxed{2.25}$$

7. For this problem, let $f(x) = (2x - 1)^{5/2}$.

(a) (5 points) Find the second Taylor polynomial, $T_2(x)$, for the function f based at $b = 1$.

$$\begin{aligned} f(x) &= (2x-1)^{5/2} & f(1) &= 1 \\ f'(x) &= 5(2x-1)^{3/2} & f'(1) &= 5 \\ f''(x) &= 15(2x-1)^{1/2} & f''(1) &= 15 \end{aligned}$$

$$T_2(x) = 1 + 5(x-1) + \frac{15}{2}(x-1)^2$$

(b) (4 points) Use $T_2(x)$ to approximate the value of $1.04^{5/2}$.

$$\begin{aligned} 1.04^{5/2} &= (2x-1)^{5/2} \\ &\downarrow \\ 1.04 &= 2x-1 \\ x &= 1.02 \\ 1.04^{5/2} &= f(1.02) \approx T_2(1.02) \\ &= 1 + 5(.02) + \frac{15}{2}(.02)^2 = 1.103 \end{aligned}$$

$$1.04^{5/2} \approx 1.103$$

(c) (5 points) Use Taylor's inequality to find an upper bound for the error in your approximation in part (b).

$$f'''(x) = \frac{15}{\sqrt{2x-1}} \quad \text{on } [1, 1.02], \text{ this is greatest when } x=1$$

$$M = 15$$

$$|f(x) - T_2(x)| \leq \frac{1}{6} \cdot 15 (.02)^3 = 0.00002$$

(larger bounds ok w/ justification)

$$\text{Upper bound: } 0.00002$$

8. For this problem, let $f(x) = \frac{x^4}{1+3x^6} + x \sin(2x^3)$

(a) (6 points) Give the Taylor series for f based at $b = 0$ using one sigma sign.

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$$

$\downarrow x \rightarrow -3x^6$

$$\frac{1}{1+3x^6} = \sum_{k=0}^{\infty} (-1)^k 3^k x^{6k}$$

$$\frac{x^4}{1+3x^6} = \sum_{k=0}^{\infty} (-1)^k 3^k x^{6k+4}$$

$$\sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$$

$\downarrow x \rightarrow 2x^3$

$$\sin(2x^3) = \sum_{k=0}^{\infty} \frac{(-1)^k 2^{2k+1} x^{6k+3}}{(2k+1)!}$$

$$x \sin(2x^3) = \sum_{k=0}^{\infty} \frac{(-1)^k 2^{2k+1} x^{6k+4}}{(2k+1)!}$$

Taylor series:

$$\sum_{k=0}^{\infty} (-1)^k \left(3^k + \frac{2^{2k+1}}{(2k+1)!} \right) x^{6k+4}$$

(b) (3 points) Find the largest open interval on which this Taylor series converges.

$$\sum_{k=0}^{\infty} x^k \text{ converges for } -1 < x < 1$$

$\downarrow x \rightarrow -3x^6$

$$-1 < -3x^6 < 1$$

$$-\frac{1}{3} < x^6 < \frac{1}{3}$$

Other transformations are irrelevant

(T. series for $\sin x$ converges everywhere)

Interval:

$$\left(-\sqrt[6]{\frac{1}{3}}, \sqrt[6]{\frac{1}{3}} \right)$$

(c) (5 points) Find $f^{(1000)}(0)$ (i.e. the 1000th derivative of f at 0).

Need x^{1000} term of $\sum_{k=0}^{\infty} (-1)^k \left(3^k + \frac{2^{2k+1}}{(2k+1)!} \right) x^{6k+4} \rightarrow \begin{matrix} 6k+4=1000 \\ 6k=996 \\ k=166 \end{matrix}$

It's $\left(3^{166} + \frac{2^{333}}{333!} \right) x^{1000}$.

By def'n of T. series, $3^{166} + \frac{2^{333}}{333!} = \frac{f^{(1000)}(0)}{1000!}$

$$f^{(1000)}(0) = 1000! \left(3^{166} + \frac{2^{333}}{333!} \right)$$